

Mawlana Bhashani Science and Technology University
Department of Information and Communication Technology
3rd Year 2nd Semester B. Sc. (Engg.) Final Examination 2025
Course Title: Computer Architecture and Organization **Course Code: ICT 3207**
Full Marks: 70 **Time: 3 Hours**

Answer any five from the following questions

1. (a) Define and Compare Computer Architecture and Computer Organization. 2
 (b) Draw and explain the internal organization of a CPU, showing ALU, registers, and control unit. 4
 (c) Explain Flynn's Taxonomy of computer architectures with suitable examples. 4
 (d) Suppose, a healthcare monitoring system requires real-time data processing, low latency, and energy-efficient computation. The system must support parallel processing and frequent data access from memory. Design a suitable computer architecture and organization for this system and justify your design choices. 4
2. (a) Draw the basic architecture of computer and explain its components in details. 5
 (b) Explain the flowchart and steps of central processing unit (CPU) operation. 4
 (c) A computer uses 12-bit memory addresses. Calculate the total number of addressable memory locations. 495 5
3. (a) Define RISC and CISC architecture of a computer. 4
 (b) Which type of instruction set architecture is used in modern processors such as Intel Core i7 — RISC, CISC, or a combination of both? Explain how it works. 4
 (c) Match the processors in Set A = {Intel Core 2 Duo, Core i3, i5, i7, i9} with their correct performance categories in Set B = {Entry-level, Gaming & creative performance, Enthusiast & high-end performance, Balanced performance}. 4
 (d) If the main memory access time is 100 ns, cache access time is 50 ns, cache hit rate is 80%, then what is average memory access time to read from memory? 600 2
4. (a) Derive the equation of three-level cache memory by using probability flowchart. 6
 (b) Compare and contrast Sequential, Direct, and Random access methods in computer memory systems. Your answer should detail the access mechanism, variability of access time, and provide a real-world example for each. 4
 (c) Let's say you're uploading a 4 GB video file from your laptop to cloud storage (like Google Drive or AWS S3) with Transfer rate (R) = 100 Mbps, Average access latency (TA) = 100 ms for Time for handshake, connection setup, DNS lookup, and request acknowledgment, then calculate average time to read and write. 327.780 4
5. (a) What is Cache memory? Explain principle of cache memory with three level cache organization. 4
 (b) Draw and explain the flow chart of cache memory read operation. 4
 (c) Consider a computer system where the internal memory uses an address length of 10 bits. Calculate the total number of addressable units available in this system. 1024 3
 (d) Suppose, CPU generates 400 references. Number of miss in level-1 cache is 40, number of miss in level-2 cache is 20. If hit time of level-1 is 1 cycle and hit time of level-2 is 10 cycle. Find total average time if miss penalty of level-2 is 100 cycle. 7 cycle 3
6. (a) A benchmark program executes 2×10^9 instructions with a CPI of 1.5 on a 3 GHz processor. Calculate the total execution time and comment on the result. 4
 (b) Convert the following numbers: 4
 i) Convert hexadecimal number 3A₁₆ to binary ii) Represent -18 in 8-bit 2's complement form
 (c) Explain the hardware implementation of unsigned binary multiplication with flowchart to perform unsigned binary multiplication of two n-bit numbers. Then, using your flowchart, compute the product of 11 × 11 in binary, showing all intermediate steps. 6
7. (a) Explain sign magnitude representation of integers with proper examples and equations. 4
 (b) Represent -25 in 8-bit 2's complement form. 3
 (c) Using Booth's Algorithm, perform the signed binary multiplication of -3×5 . Draw the flowchart then show all steps, including the accumulator, multiplier, extra bit, and the final product. 1110001 7
8. (a) Explain is pipelining with example like $A_i \times B_i + C_i$ where $i = 1, 2, 3, \dots, 7$ for pipeline processing and pipeline registers. 7
 (b) Analyze how virtual memory enables efficient multiprogramming, and discuss the performance implications of frequent page faults. 3
 (c) Explain Data Hazards and its Handling Methods. 4